

Gibbs free energy under non-standard conditions (Ch. 16)

The relationship between ΔG° and ΔG is given by:

$$\Delta G = \Delta G^\circ + RT \ln(Q)$$

where Q is the **reaction quotient**.

For a general reaction,



Q is given by

$$Q = \frac{[C]^\gamma [D]^\delta}{[A]^\alpha [B]^\beta}$$

At equilibrium (from Ch. 16)

$\Delta G = 0$, and Q becomes the *equilibrium constant*, K :

$$0 = \Delta G^\circ + RT \ln(K)$$

Which allows the equilibrium constant to be determined from knowing ΔG° :

$$K = \exp(-\Delta G^\circ/RT)$$

K	$\ln(K)$	ΔG°	Comments
> 1	Positive	Negative	Products favored over reactants
$= 1$	0	0	Products and reactants equally favored
< 1	Negative	Positive	Reactants are favored over products

Rules for K expressions



1. Concentrations in solution are given in M, in gas phase typically in atm or bar.
2. The concentration of pure solids, pure liquids (heterogeneous equilibria), and solvents do not appear in the expressions.

EXAMPLE: *Solubility product*

3. In quoting a value for the equilibrium constant, we must specify the balanced equation and the temperature.

Le Châtelier's principle



*If an **equilibrium** is disturbed, the position of equilibrium shifts to counteract the change to reestablish an equilibrium.*

If a chemical reaction is at equilibrium and experiences a change in pressure, temperature, or concentration of products or reactants, the equilibrium shifts in the opposite direction to offset the change.

EXAMPLE:



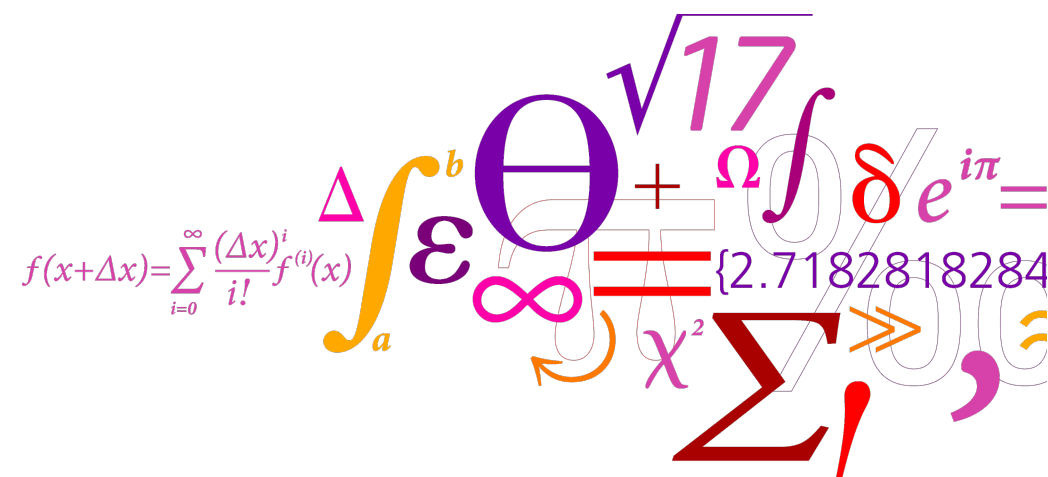
Example exercise:

Consider again the reaction $2 \text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$

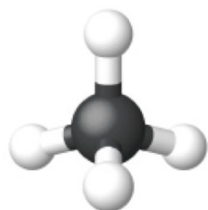
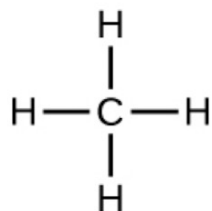
- ▶ Write down the reaction quotient.
- ▶ A 1 L container is filled with 1 atm of pure NO_2 . Given that the equilibrium constant is $K = 6.1 \text{ atm}^{-1}$ at 25°C , calculate $p(\text{NO}_2)$ and $p(\text{N}_2\text{O}_4)$ at equilibrium.
- ▶ The volume of the contained is reduced to 0.5 L using a piston. Calculate the reaction quotient before equilibrium has been established. In which direction will the equilibrium shift?
- ▶ The reaction's ΔH° is -580 kJ mol^{-1} . Using Le Chatelier's principle, would cooling (to e.g. -78°C) increase or decrease the amount of NO_2 ? Can you explain that using the concept of entropy?

Organic Chemistry

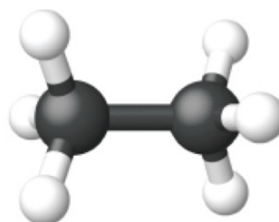
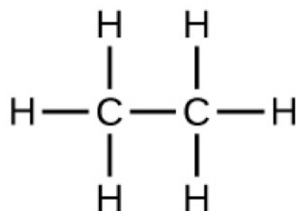
Kasper S. Pedersen



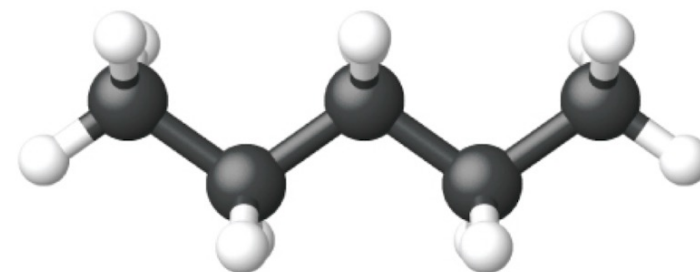
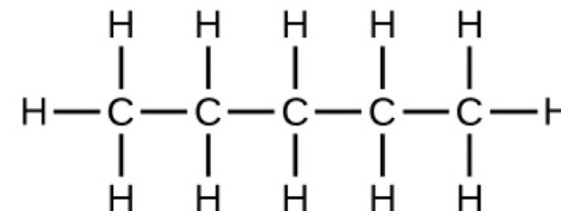
Hydrocarbons: Alkanes



methane
CH₄

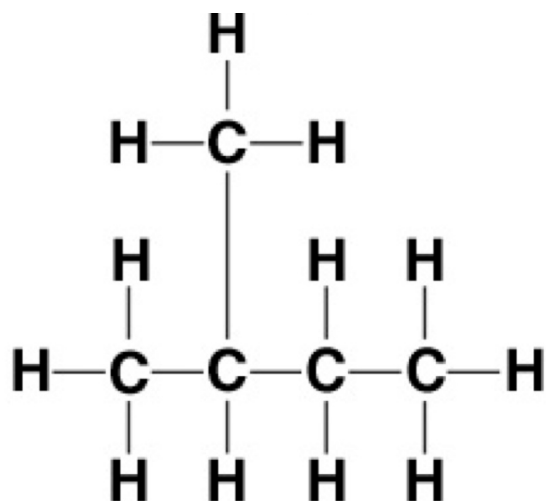


ethane
CH₃CH₃ or C₂H₆

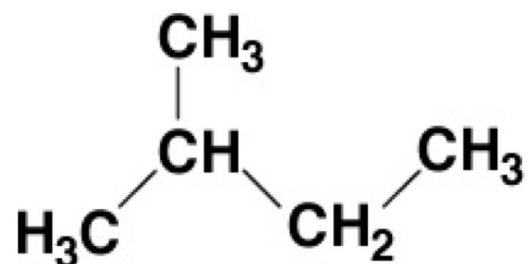


pentane
CH₃CH₂CH₂CH₂CH₃ or C₅H₁₂

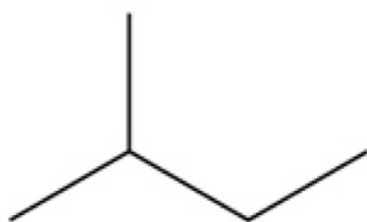
Different ways to write organic structures: 2-methylbutane



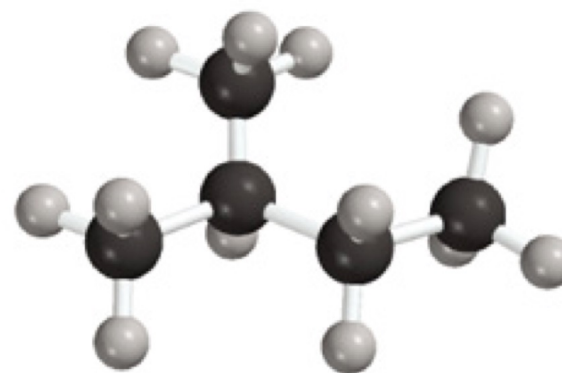
(a)



(b)



(c)

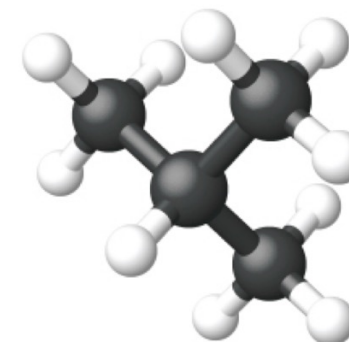
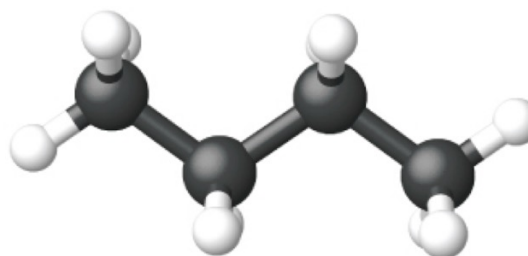
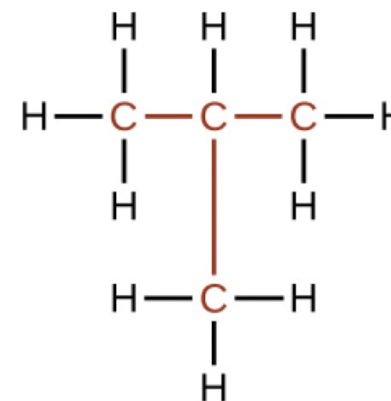
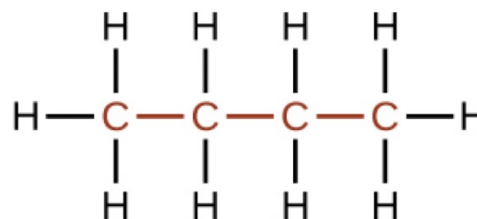


(d)

Isomers: Same net molecular formula

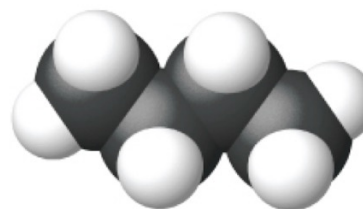
Isomers have the same molecular formula, but different structures

Because groups are placed differently, and they are therefore different molecules with different properties

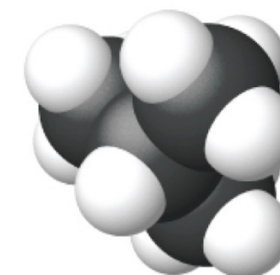


EXERCISE

Draw the isomers of C_5H_{12} and C_6H_{14} !



n-butane

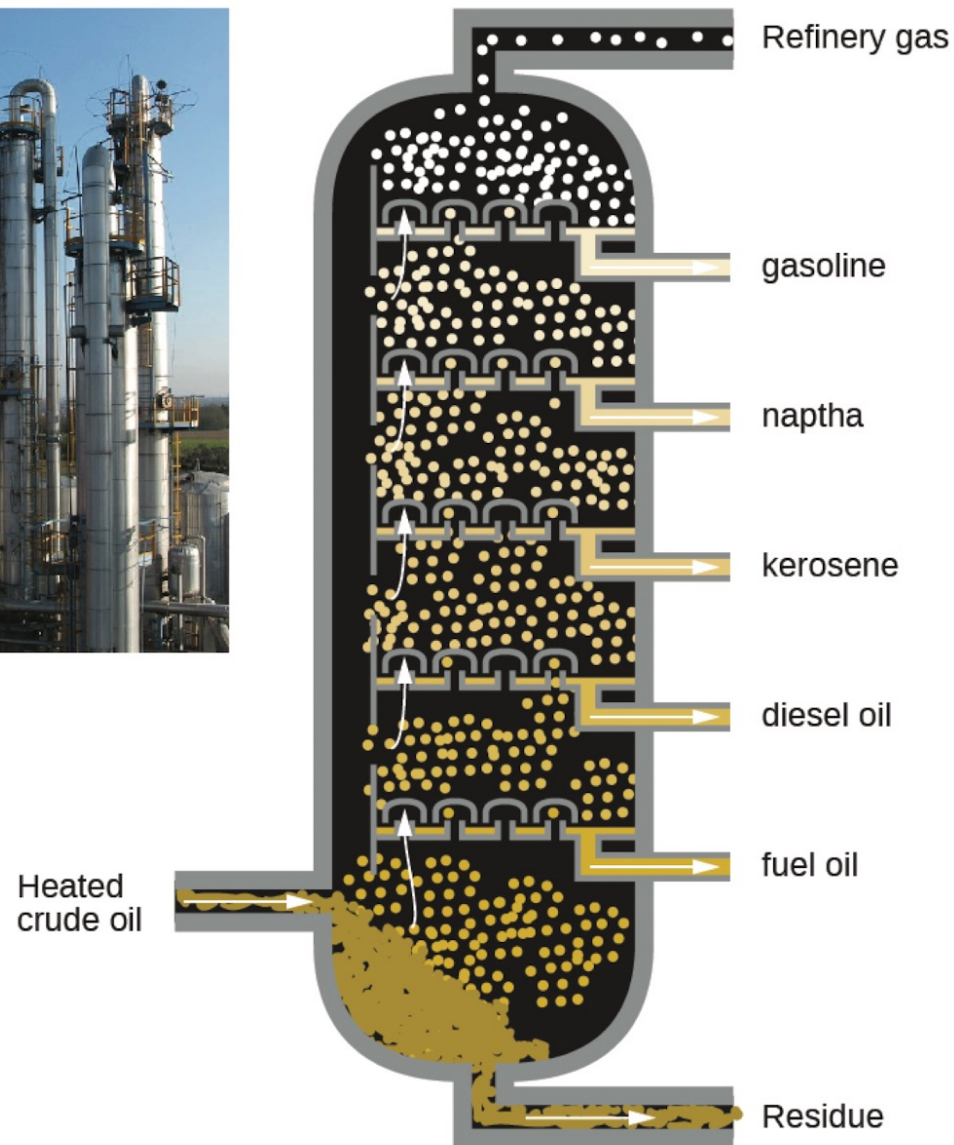


2-methylpropane

Alkanes: Physical properties

Alkane	Molecular Formula	Melting Point (°C)	Boiling Point (°C)	Phase at STP ⁴	Number of Structural Isomers
methane	CH ₄	-182.5	-161.5	gas	1
ethane	C ₂ H ₆	-183.3	-88.6	gas	1
propane	C ₃ H ₈	-187.7	-42.1	gas	1
butane	C ₄ H ₁₀	-138.3	-0.5	gas	2
pentane	C ₅ H ₁₂	-129.7	36.1	liquid	3
hexane	C ₆ H ₁₄	-95.3	68.7	liquid	5
heptane	C ₇ H ₁₆	-90.6	98.4	liquid	9
octane	C ₈ H ₁₈	-56.8	125.7	liquid	18
nonane	C ₉ H ₂₀	-53.6	150.8	liquid	35
decane	C ₁₀ H ₂₂	-29.7	174.0	liquid	75
tetradecane	C ₁₄ H ₃₀	5.9	253.5	solid	1858
octadecane	C ₁₈ H ₃₈	28.2	316.1	solid	60,523

Crude oil provides alkanes



Small molecules:
- Low boiling point
- Very volatile
- Flows easily
- Ignites easily



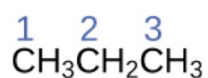
Large molecules:
- High boiling point
- Not very volatile
- Does not flow easily
- Does not ignite easily

Alkanes: Nomenclature rules

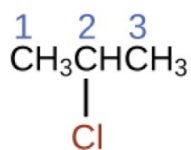


1. First identify the longest chain of carbon atoms in its structure.
A two-carbon chain is called ethane; a three-carbon chain, propane; and a four-carbon chain, butane. Longer chains are named as follows: pentane (five-carbon chain), hexane (6), heptane (7), octane (8), nonane (9), and decane (10).
2. Add prefixes to the name of the longest chain to indicate the positions and names of substituents.
Substituents are *functional groups* that replace hydrogen atoms on a chain. The position of a substituent or branch is identified by the number of the carbon atom it is bonded to in the chain. We number the carbon atoms in the chain by counting from the end of the chain nearest the substituents. Multiple substituents are named individually and placed in alphabetical order at the front of the name.

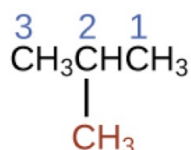
Examples



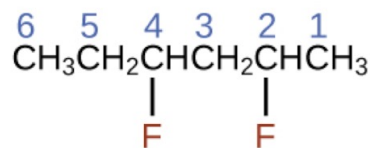
propane



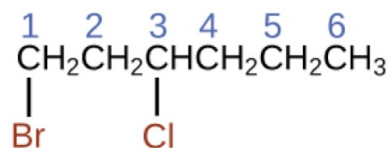
2-chloropropane



2-methylpropane



2,4-difluorohexane

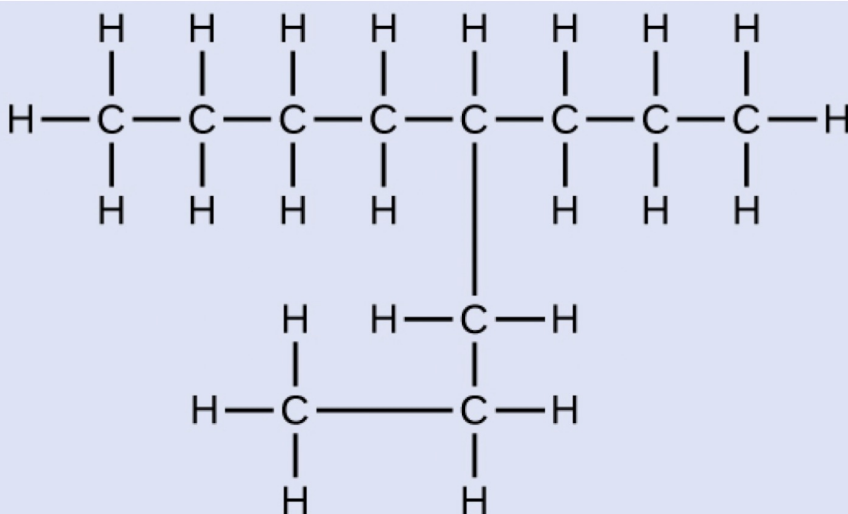


1-bromo-3-chlorohexane

Alkyl Group	Structure
methyl	$\text{CH}_3\text{—}$
ethyl	$\text{CH}_3\text{CH}_2\text{—}$
<i>n</i> -propyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{—}$
isopropyl	$\begin{array}{c} \\ \text{CH}_3\text{CHCH}_3 \end{array}$
<i>n</i> -butyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{—}$
sec-butyl	$\begin{array}{c} \\ \text{CH}_3\text{CH}_2\text{CHCH}_3 \end{array}$
isobutyl	$\begin{array}{c} \text{CH}_3\text{CHCH}_2\text{—} \\ \\ \text{CH}_3 \end{array}$
<i>tert</i> -butyl	$\begin{array}{c} \\ \text{CH}_3\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$

EXERCISE

What is the name of the alkane:



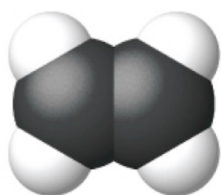
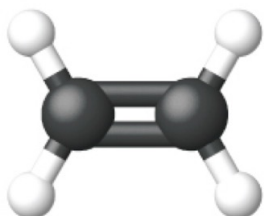
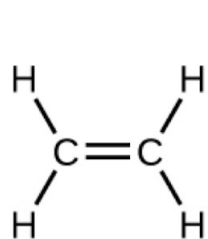
Draw the alkanes:

4-ethyl-3-methyldecane

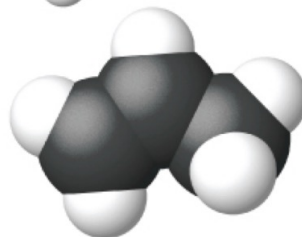
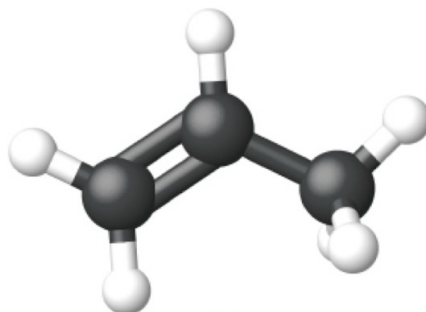
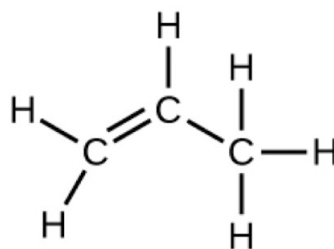
2-butyl-2-methylbutane

Alkyl Group	Structure
methyl	$\text{CH}_3\text{—}$
ethyl	$\text{CH}_3\text{CH}_2\text{—}$
<i>n</i> -propyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{—}$
isopropyl	$\begin{array}{c} \\ \text{CH}_3\text{CHCH}_3 \end{array}$
<i>n</i> -butyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{—}$
sec-butyl	$\begin{array}{c} \\ \text{CH}_3\text{CH}_2\text{CHCH}_3 \end{array}$
isobutyl	$\begin{array}{c} \text{CH}_3\text{CHCH}_2\text{—} \\ \\ \text{CH}_3 \end{array}$
<i>tert</i> -butyl	$\begin{array}{c} \\ \text{CH}_3\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$

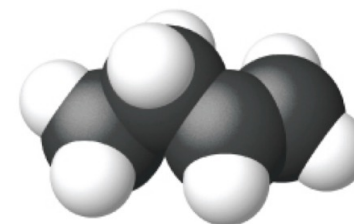
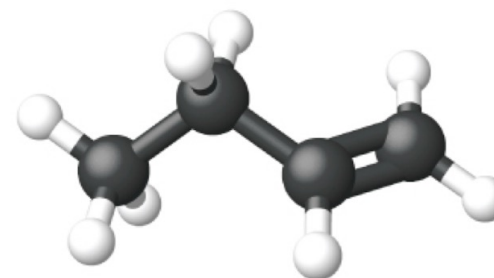
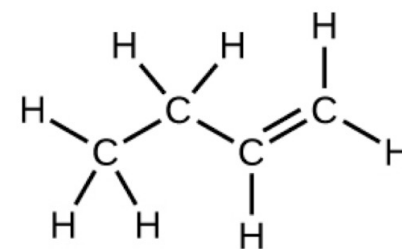
Unsaturated hydrocarbons: Alkenes



ethene

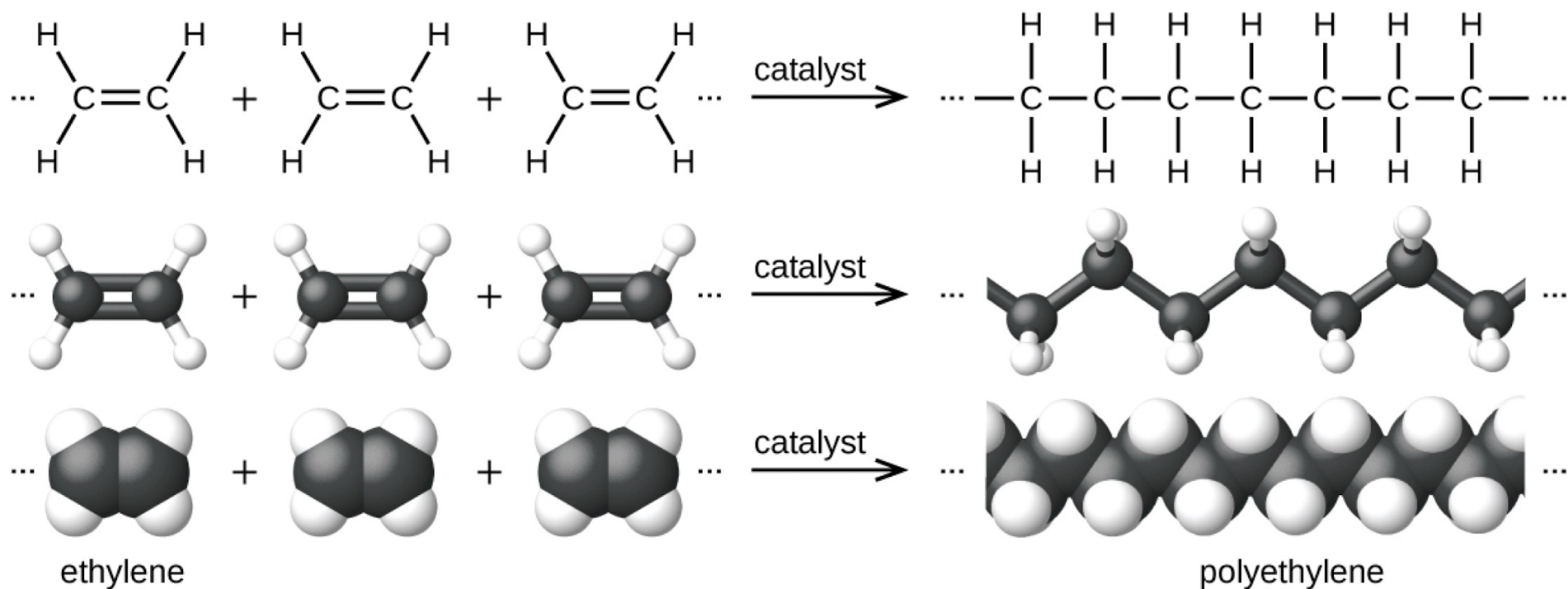


propene

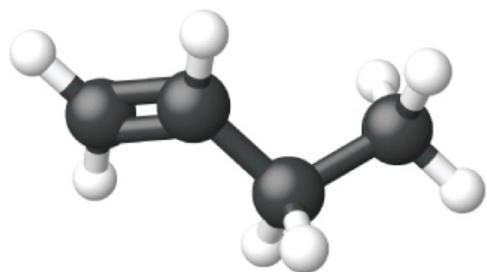
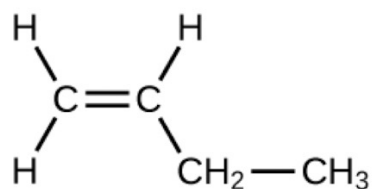


1-butene

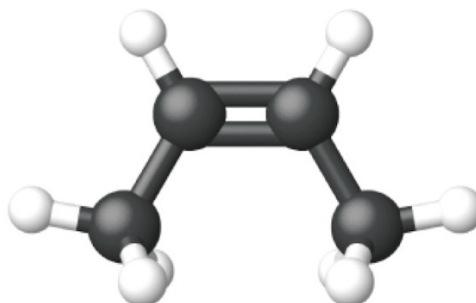
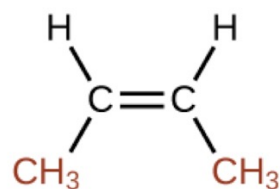
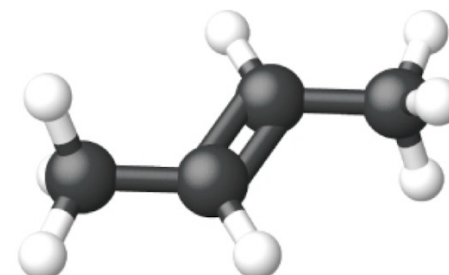
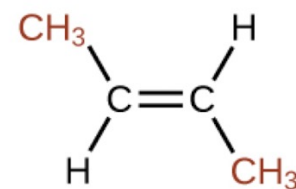
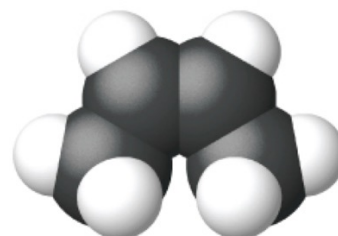
Unsaturated hydrocarbons: Alkenes



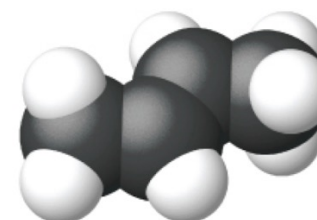
Unsaturated hydrocarbons: Alkenes



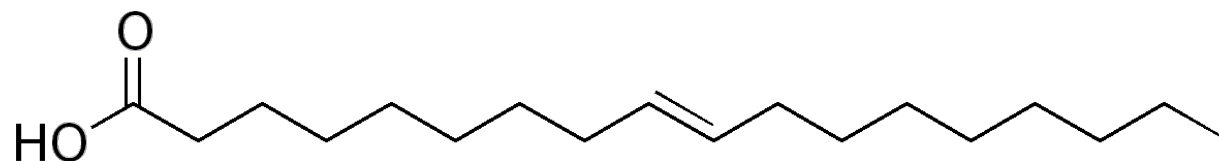
1-butene


cis isomer

trans isomer


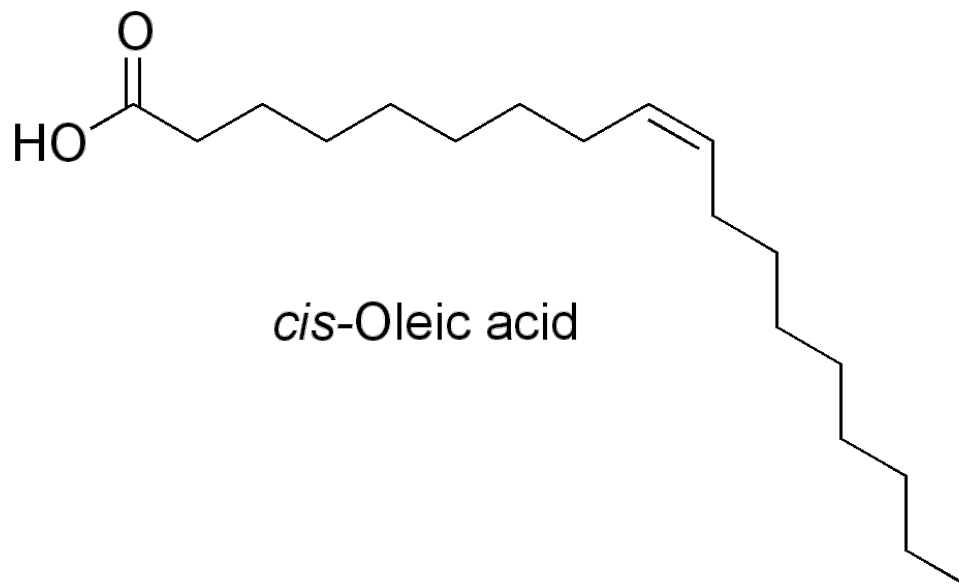
2-butene



Structures: cis/trans isomers



trans-Oleic acid



cis-Oleic acid

Alkynes

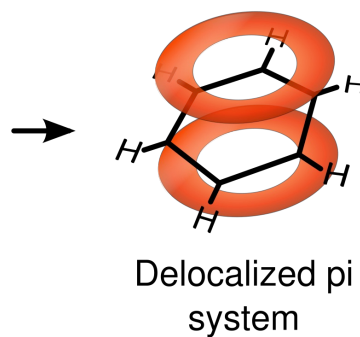
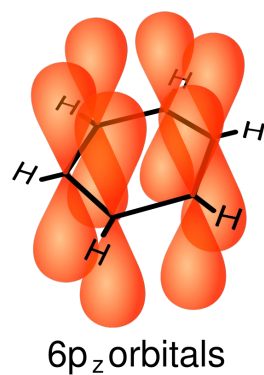
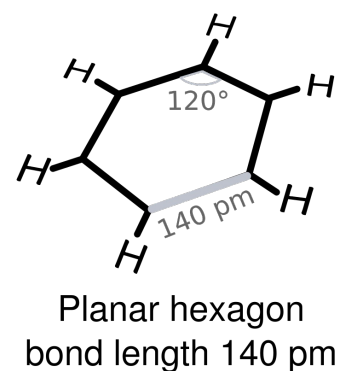
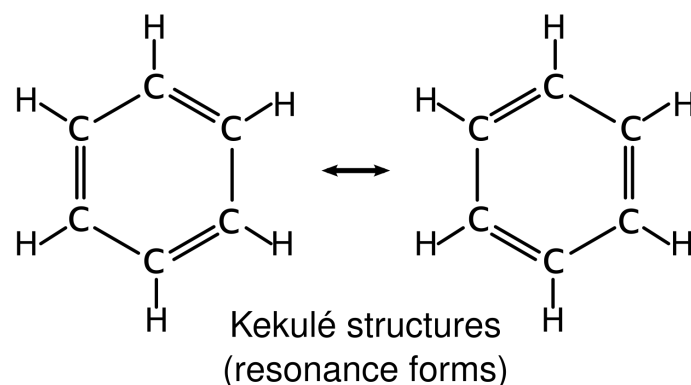
EXERCISE

Draw the Lewis structure of C_2H_2



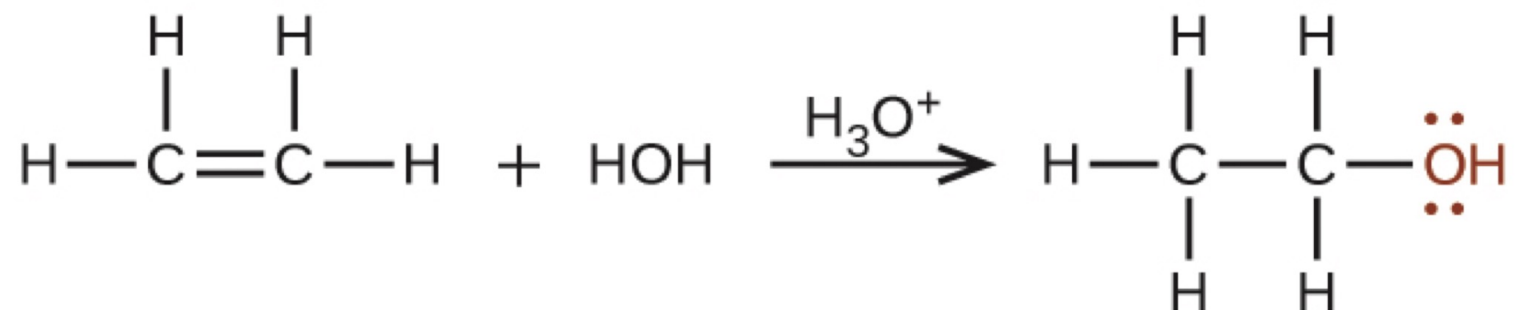
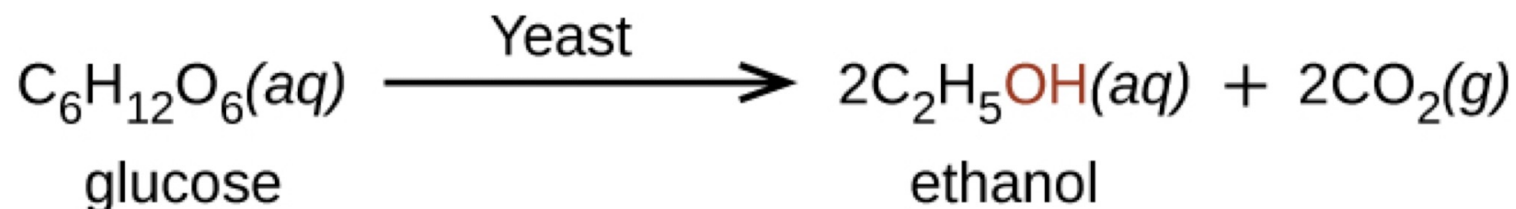
Aromatic hydrocarbons: Benzene

Benzene = Planar 6-membered ring with one hydrogen on each carbon.



Alcohols

Alcohols in addition to the main alkyl (R) group contain an **OH (hydroxyl) group**.



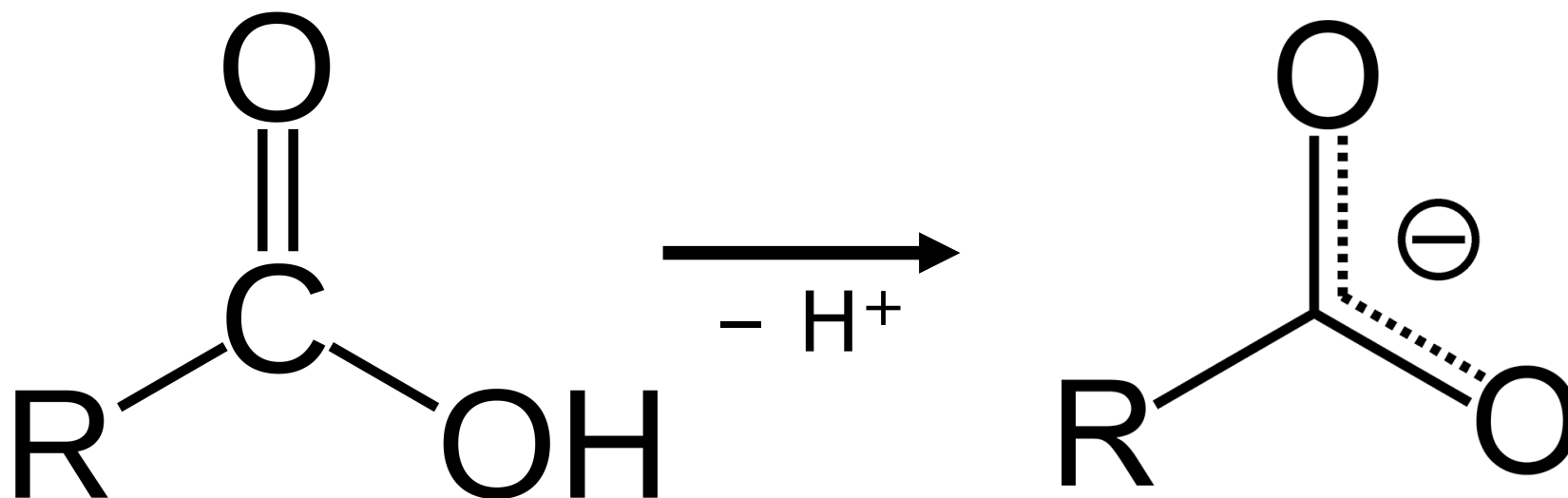
Overview of functional groups

Compound Name	Structure of Compound and Functional Group (red)
alkene	$\text{C}=\text{C}$
alkyne	$\text{C}\equiv\text{C}$
alcohol	$\text{R}-\ddot{\text{O}}-\text{H}$
ether	$\text{R}-\ddot{\text{O}}-\text{R}'$
aldehyde	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\text{H} \end{array}$
ketone	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\text{R}' \end{array}$
carboxylic acid	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{O}}-\text{H} \end{array}$

ester	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{O}}-\text{R}' \end{array}$
amine	$\begin{array}{ccc} \text{R}-\ddot{\text{N}}-\text{H} & \text{R}-\ddot{\text{N}}-\text{H} & \text{R}-\ddot{\text{N}}-\text{R}'' \\ & & \\ \text{H} & \text{R}' & \text{R}' \end{array}$
amide	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{N}}-\text{R}' \\ \\ \text{H} \end{array}$

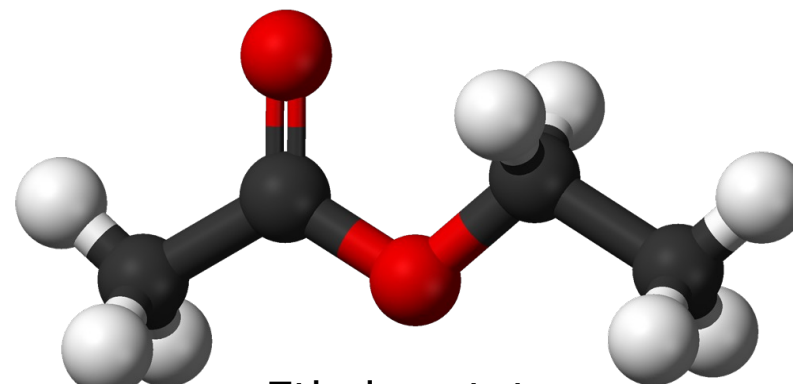
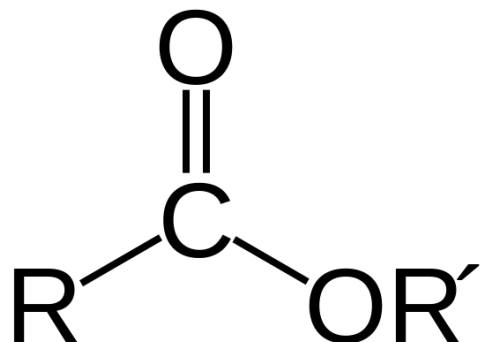
Carboxylic acids

Carboxylic acids contain a **COOH-group**.

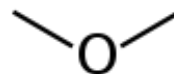
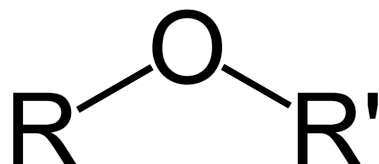


Esters and ethers

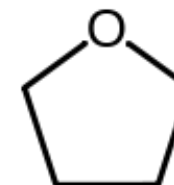
Esters contain the **COO-group** (one double bond C=O, the other O single bonded to C). Ethers contain C-O-C (single bonded)



Ethyl acetate



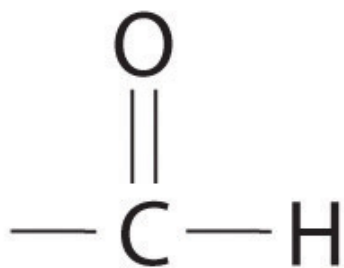
Dimethyl ether



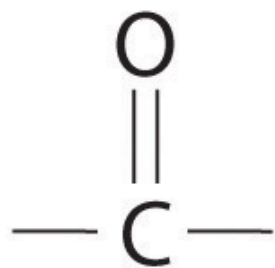
Tetrahydrofuran

Aldehydes and ketones

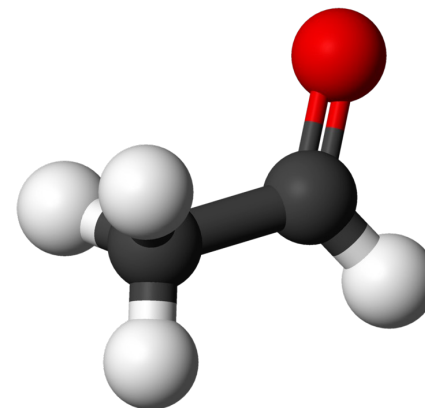
Contain a CO group:



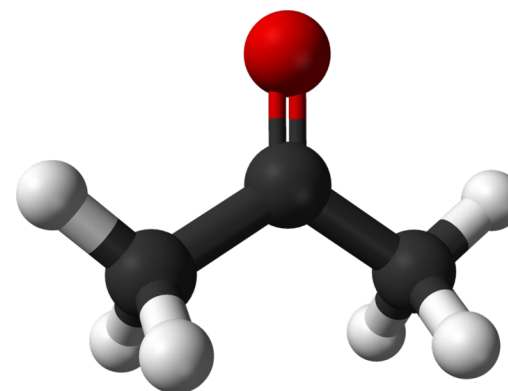
Aldehyde



Ketone

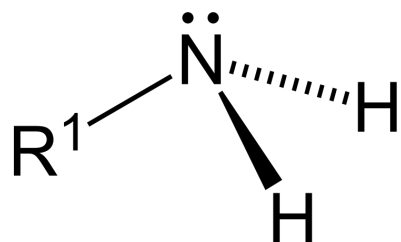


Acetaldehyde

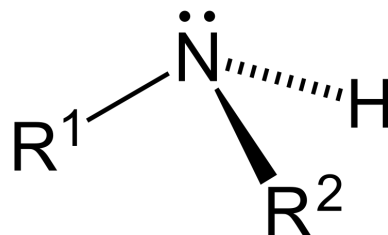


Acetone

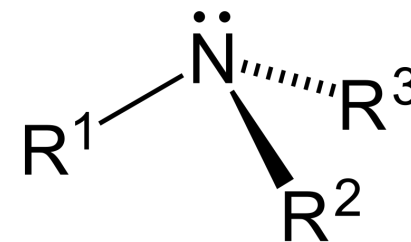
Amines



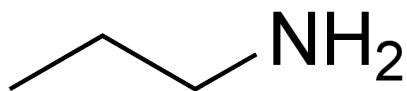
Primary amine



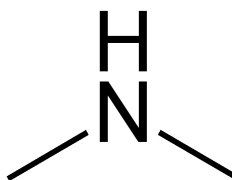
Secondary amine



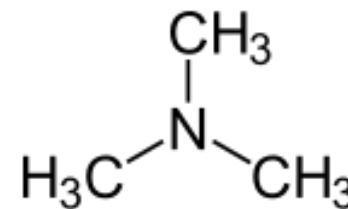
Tertiary amine



Propyl amine



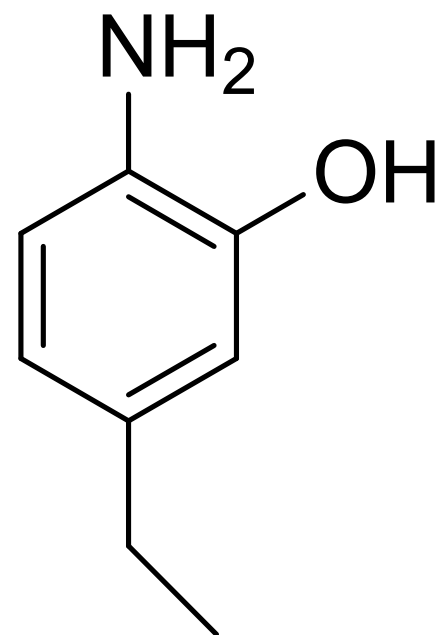
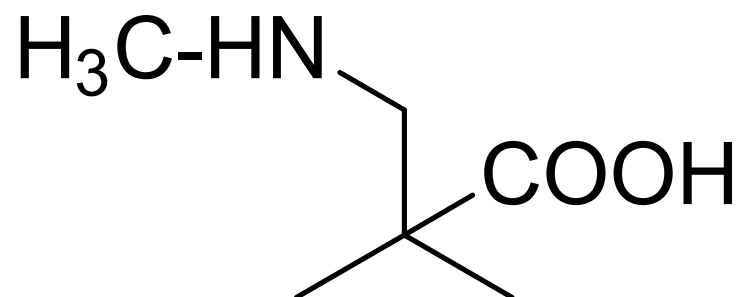
Dimethyl amine



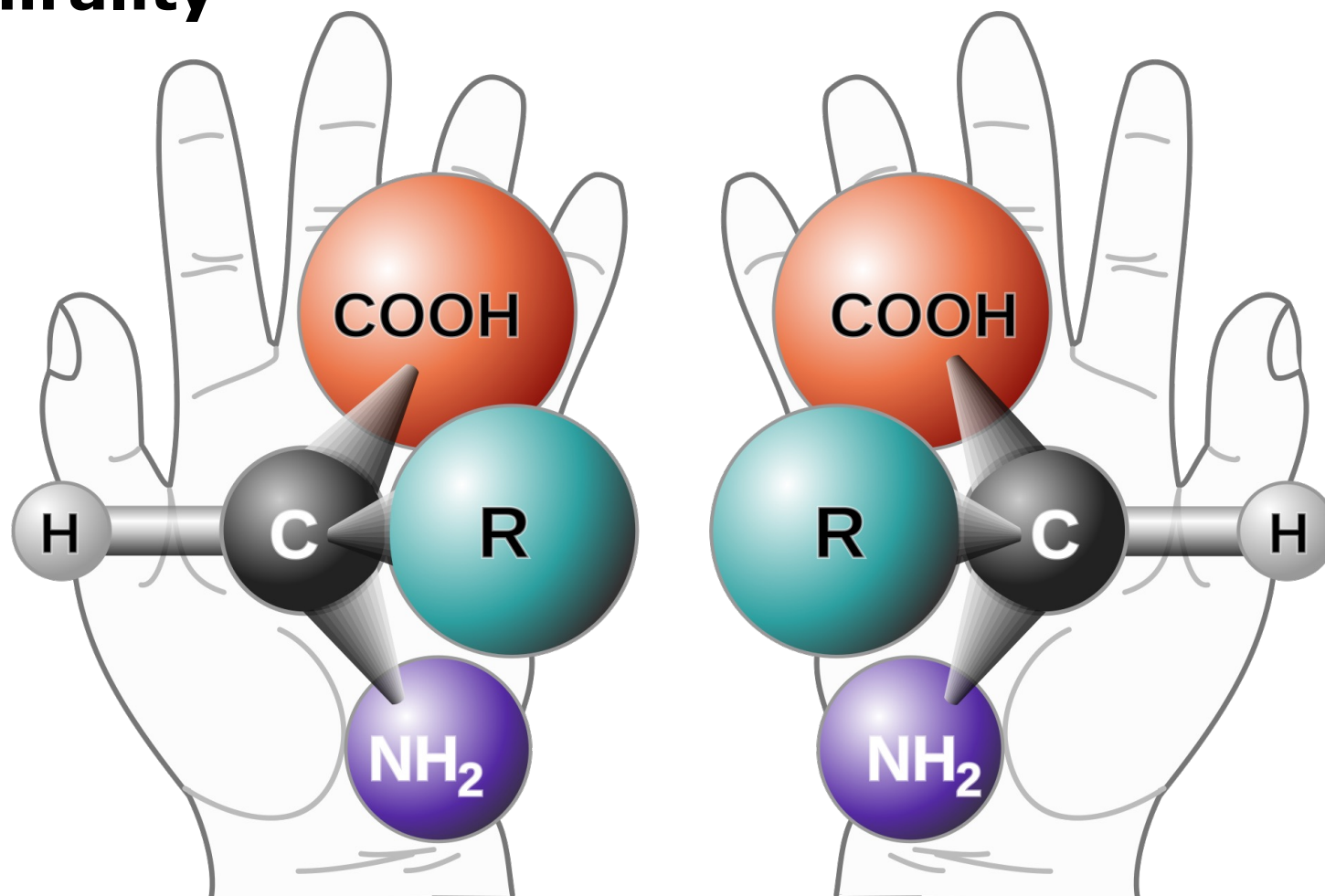
Trimethyl amine

Task

Identify functional groups in the structures below

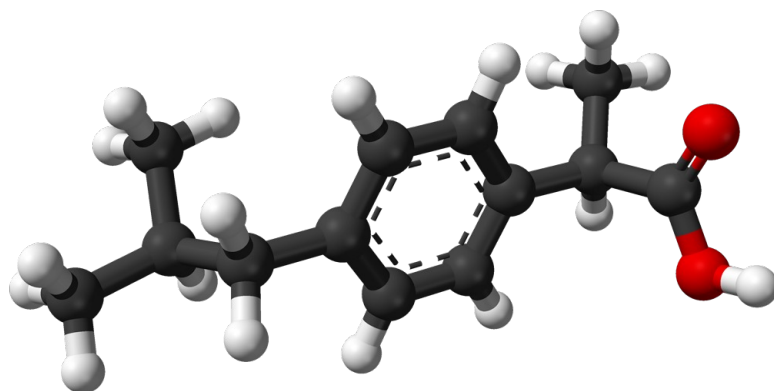


Chirality

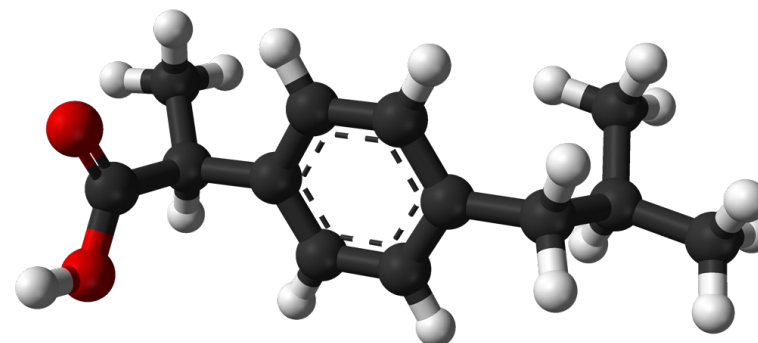


Chirality

Chiral molecules have C atoms with four different groups attached. Switching two of these will create an **enantiomeric pair**. The asymmetric atom is labelled with *.



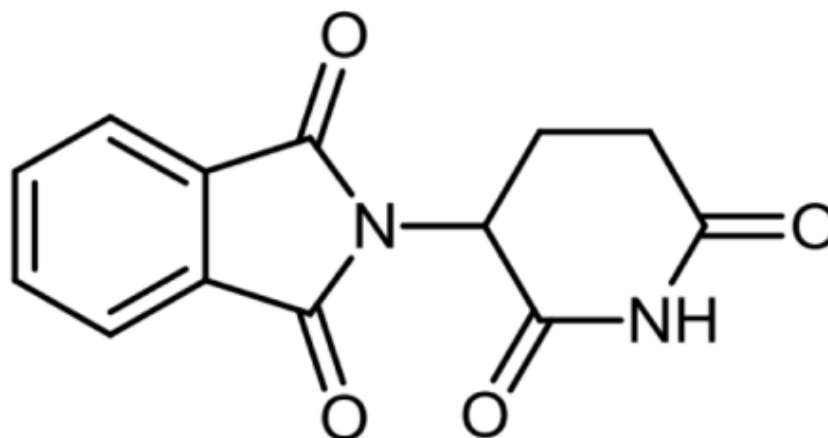
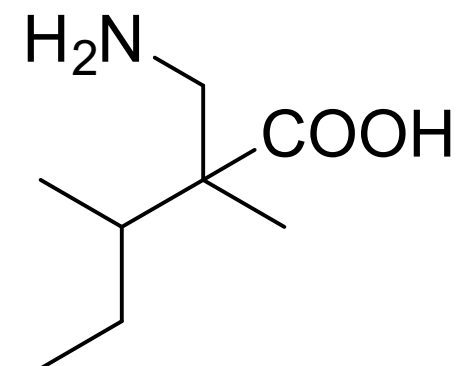
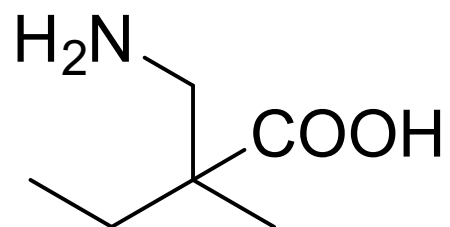
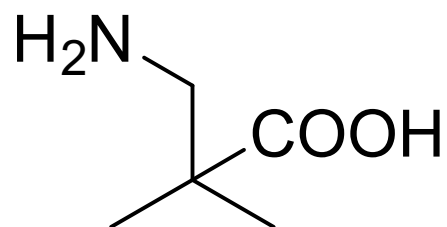
R-ibuprofen



S-ibuprofen (active form)

Exercise

Find chiral centers in the structures below:



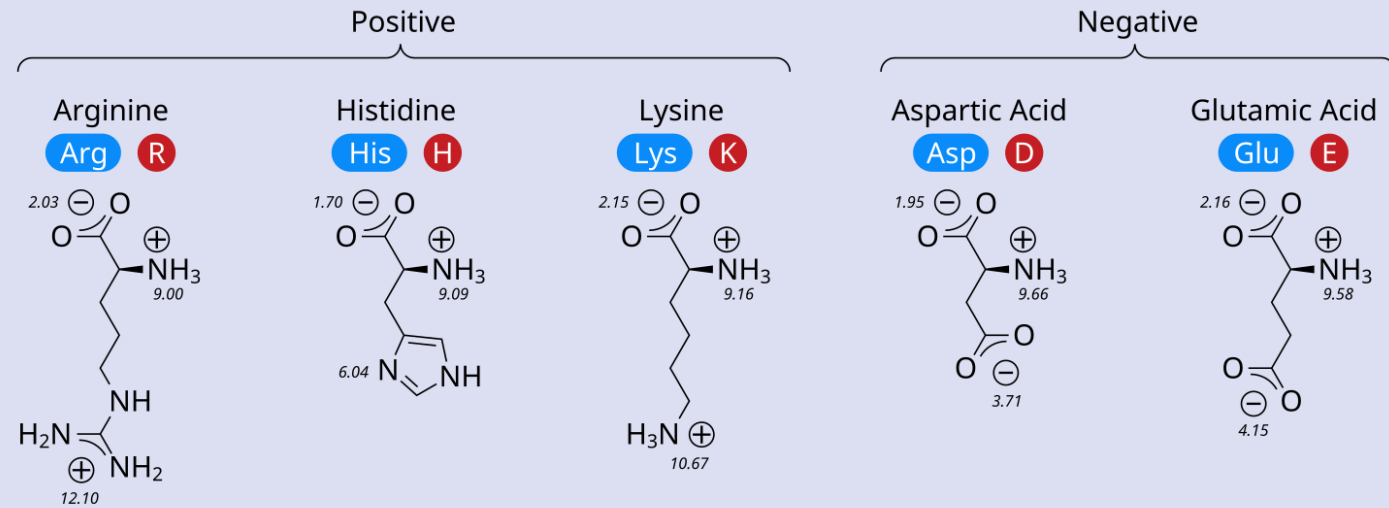
TWENTY-ONE PROTEINOGENIC α -AMINO ACIDS

Side chain charge
at physiological
pH 7.4

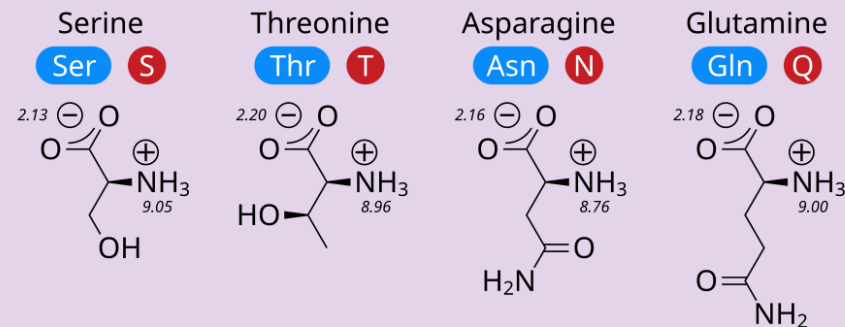
pK_a values shown
italicized

⊕ Positive
⊖ Negative

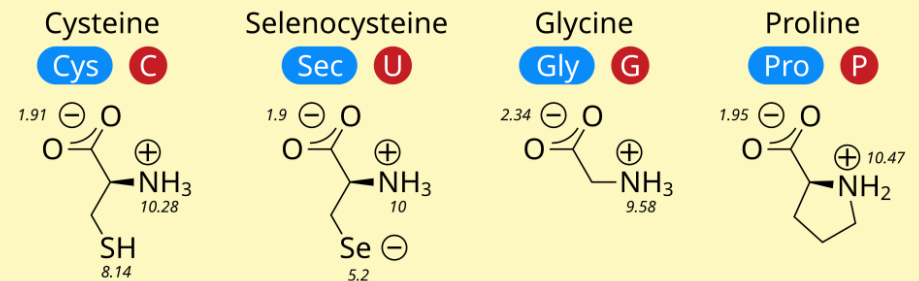
A. Amino Acids with Electrically Charged Side Chains



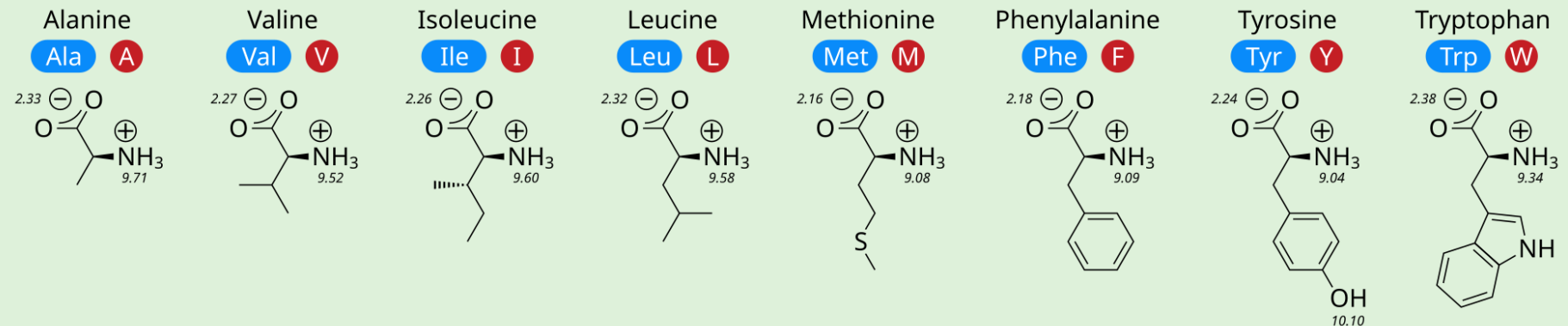
B. Amino Acids with Polar Uncharged Side Chains



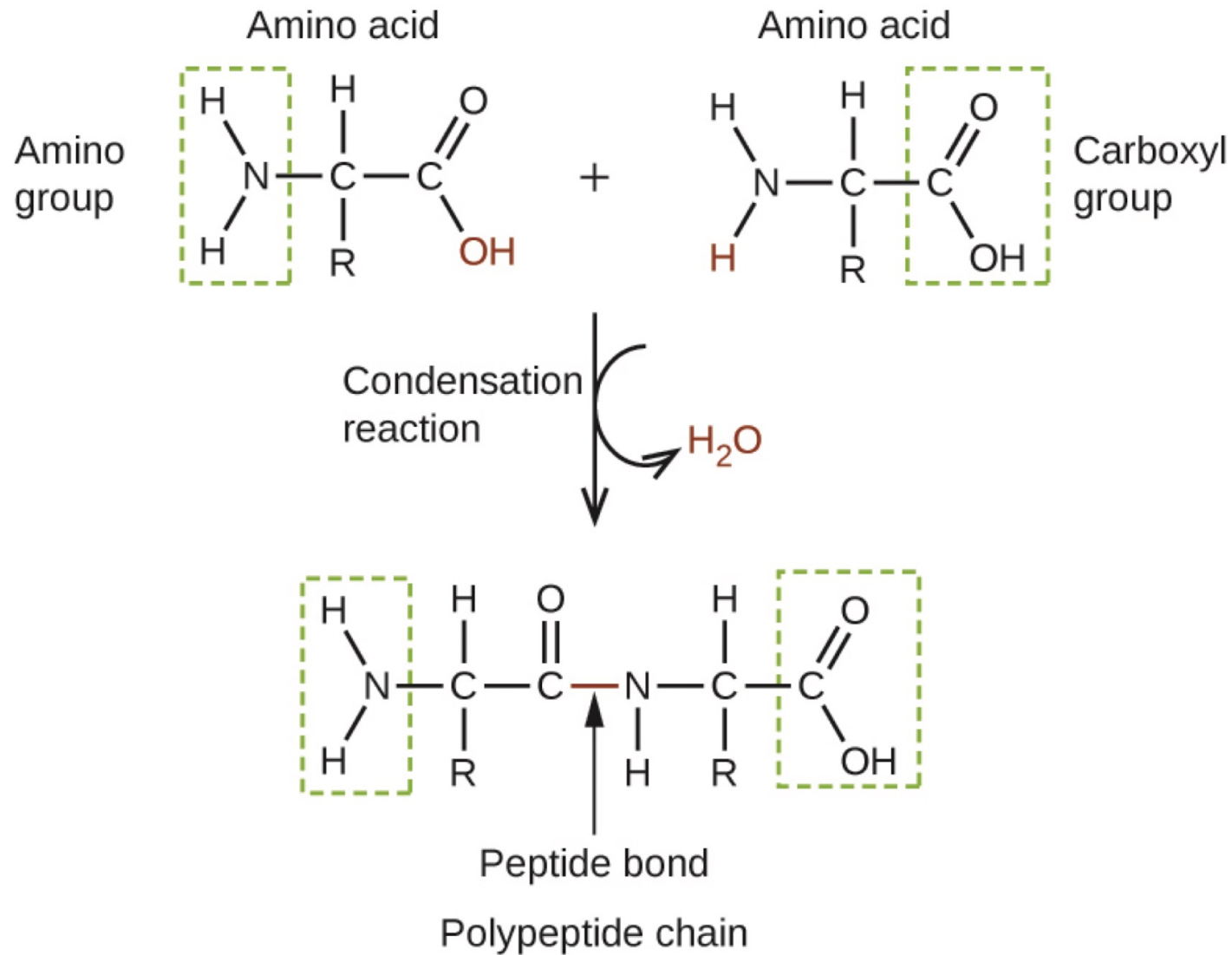
C. Special Cases



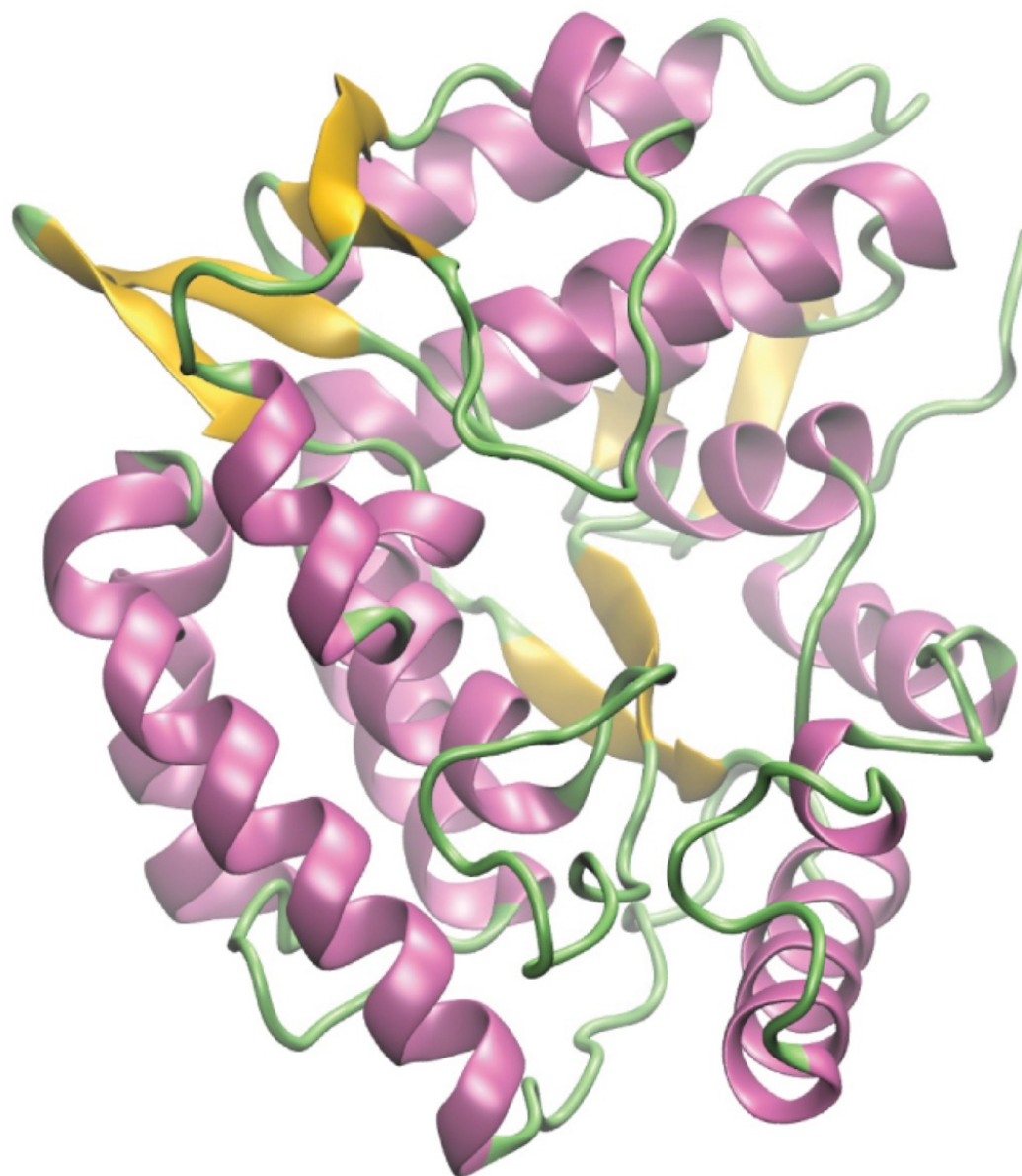
D. Amino Acids with Hydrophobic Side Chains



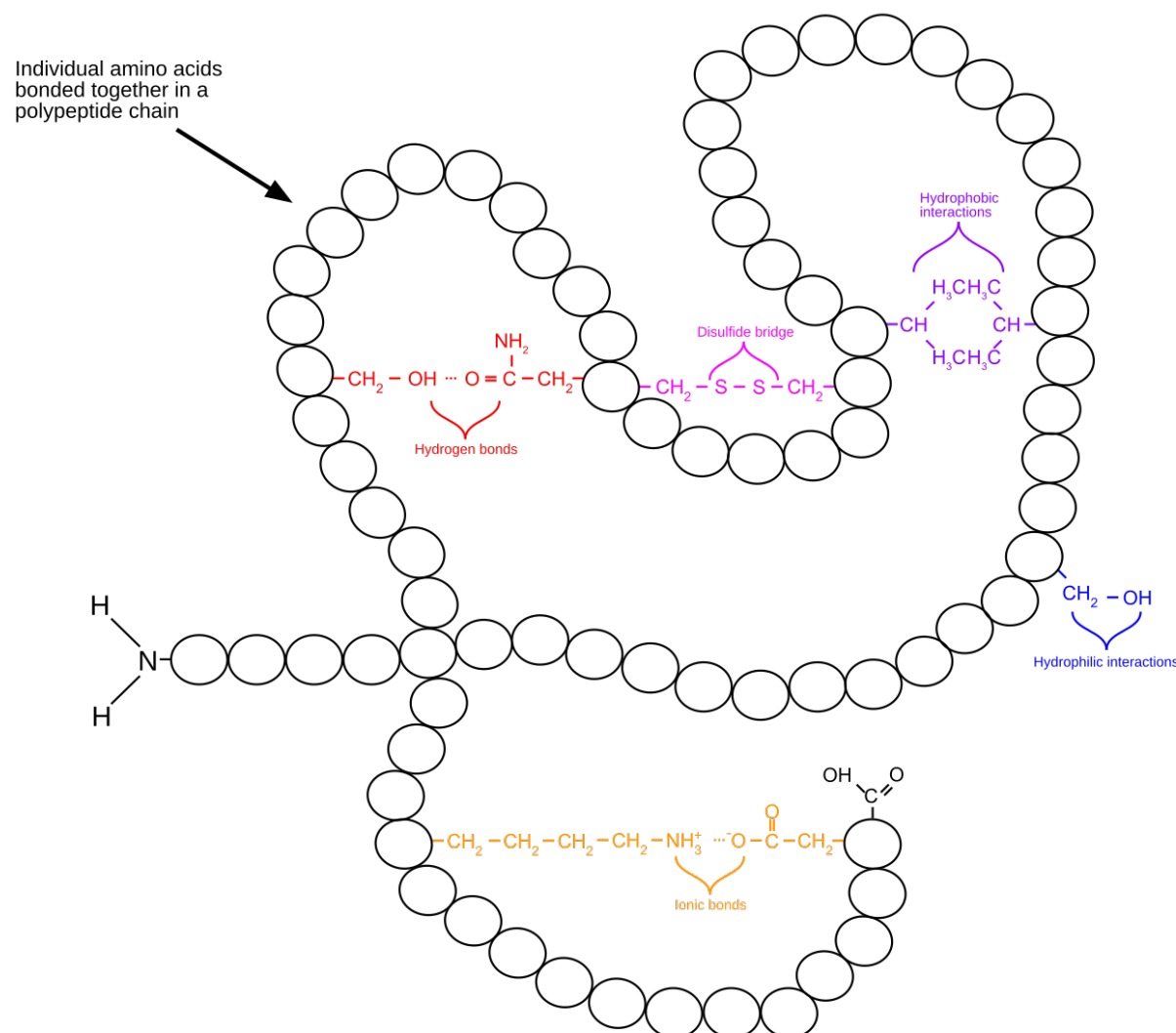
Amino acids form peptides and proteins



Protein structures



Protein structures



Bond types

Hydrophobic interactions:

These amino acids orient themselves towards the center of the polypeptide to avoid the water

Disulfide bridge: Cysteine forms a bond with another cysteine through its side group

Hydrogen bonds: Polar side groups on the amino acids form bonds with other polar side groups

Hydrophilic interactions: These amino acids orient themselves outward to be close to the water

Ionic bonds: Positively charged side groups bond to negatively charged ones